

Topics of today's class

Last class:

- Chapter 5
 - Introduce the government.
 - Construct closed-economy one-period macroeconomic model.
 - Economic efficiency and Pareto optimality.

Today's class:

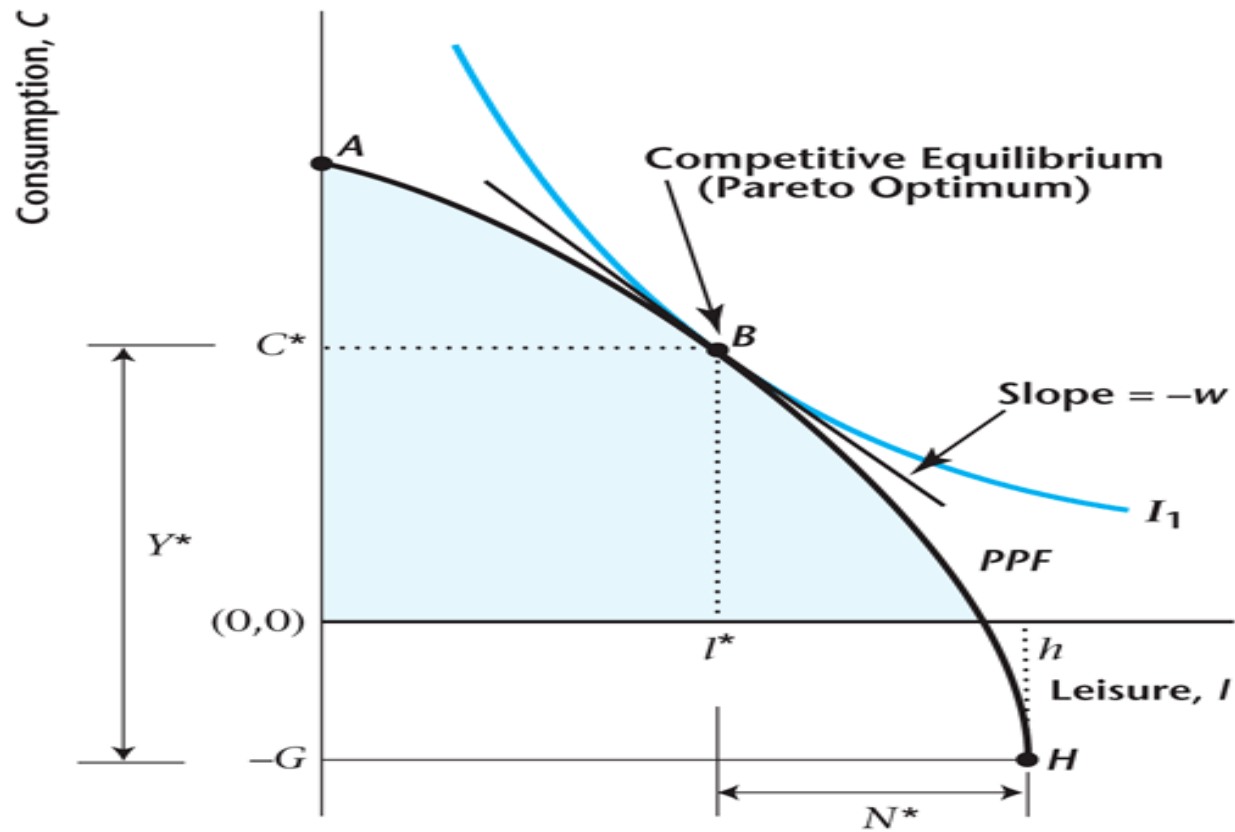
- Chapter 5
 - Increases in government spending and total factor productivity.
 - Consider a distortionary tax on wage income and study the Laffer curve.

Chapter 5

A Closed Economy One-Period Macroeconomic Model

Figure 5.4

Pareto Optimality



American Recovery and Reinvestment Act of 2009

- \$787 billion in changes in the federal government's budget (5.5% of 2008 GDP).
- \$290 billion in changes in G (2.0% of 2008 GDP).
- Our model says: effect on GDP will be less than \$290 billion, because of crowding out.
- Keynesians say: economy is operating inside the PPF, and the increase in GDP will be more than \$290 billion.

Figure 5.13

Government Expenditures as a Percentage of GDP

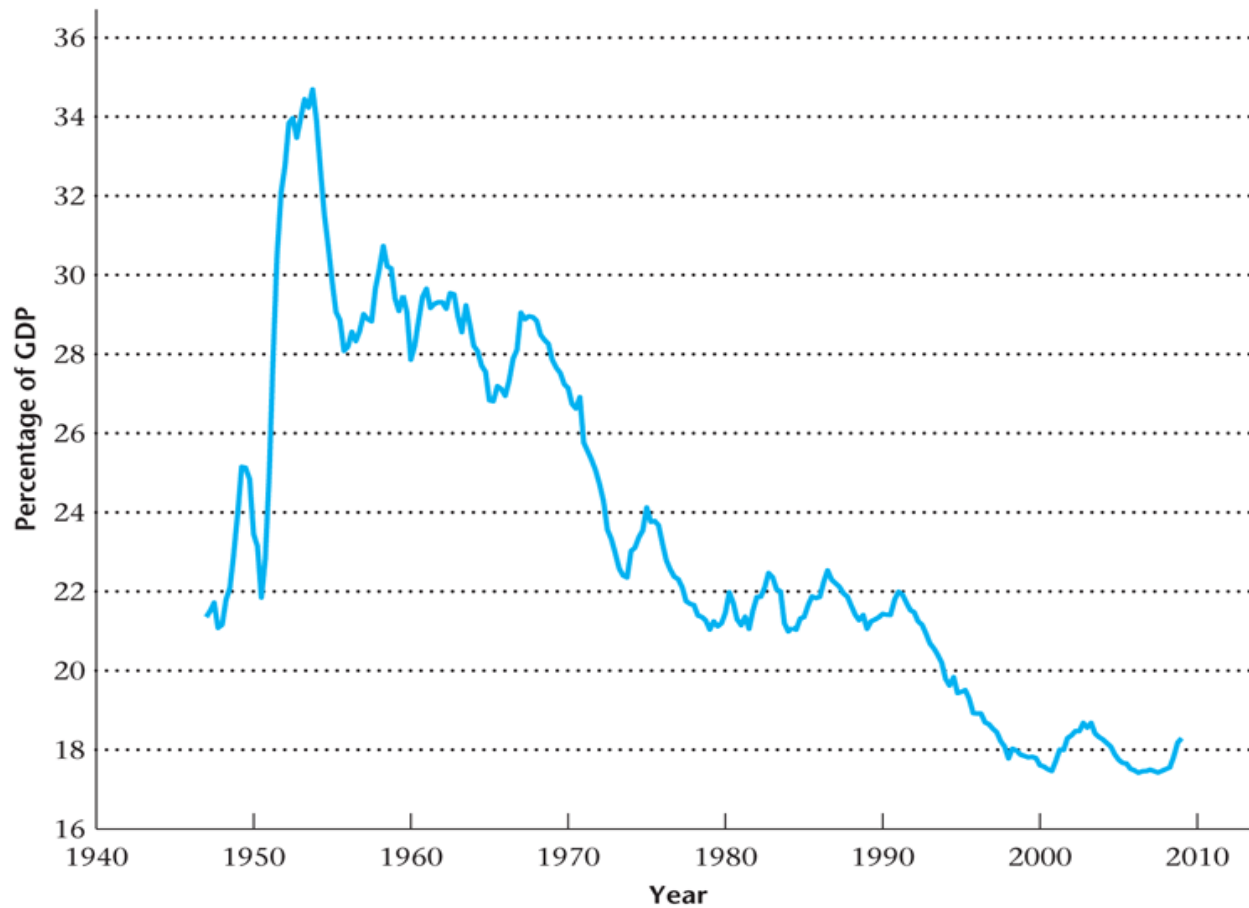
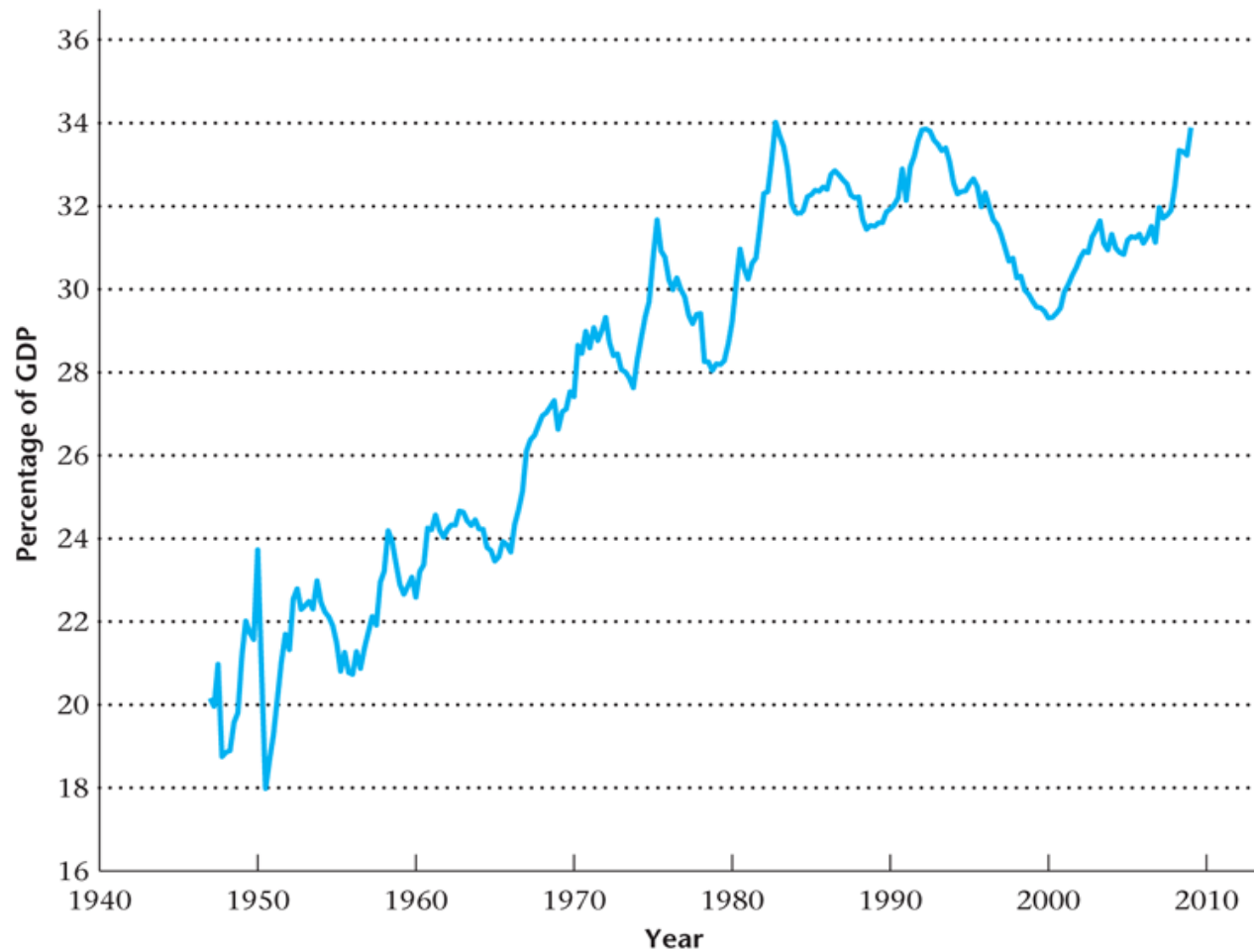


Figure 5.14

Total Government Outlays as a Percentage of GDP



American Recovery and Reinvestment Act of 2009

- In the discussions of fiscal stimulus, the size of the multiplier is a source of some controversy.
 - Obama administration suggests ≈ 1.5
 - Barro suggests ≈ 0
- Our model says: effect on GDP < 1 because of crowding out.
- This isn't the best framework for current issues, as there's no unemployment or idle resources. These are the main rationale for the fiscal stimulus.

Increase in TFP

- An increase in TFP
 - Example: technological innovation
- 2 direct effects:
 - Can produce more with the same amount of labor.
 - Production Possibility Frontier is steeper.

Figure 5.8

Increase in Total Factor Productivity

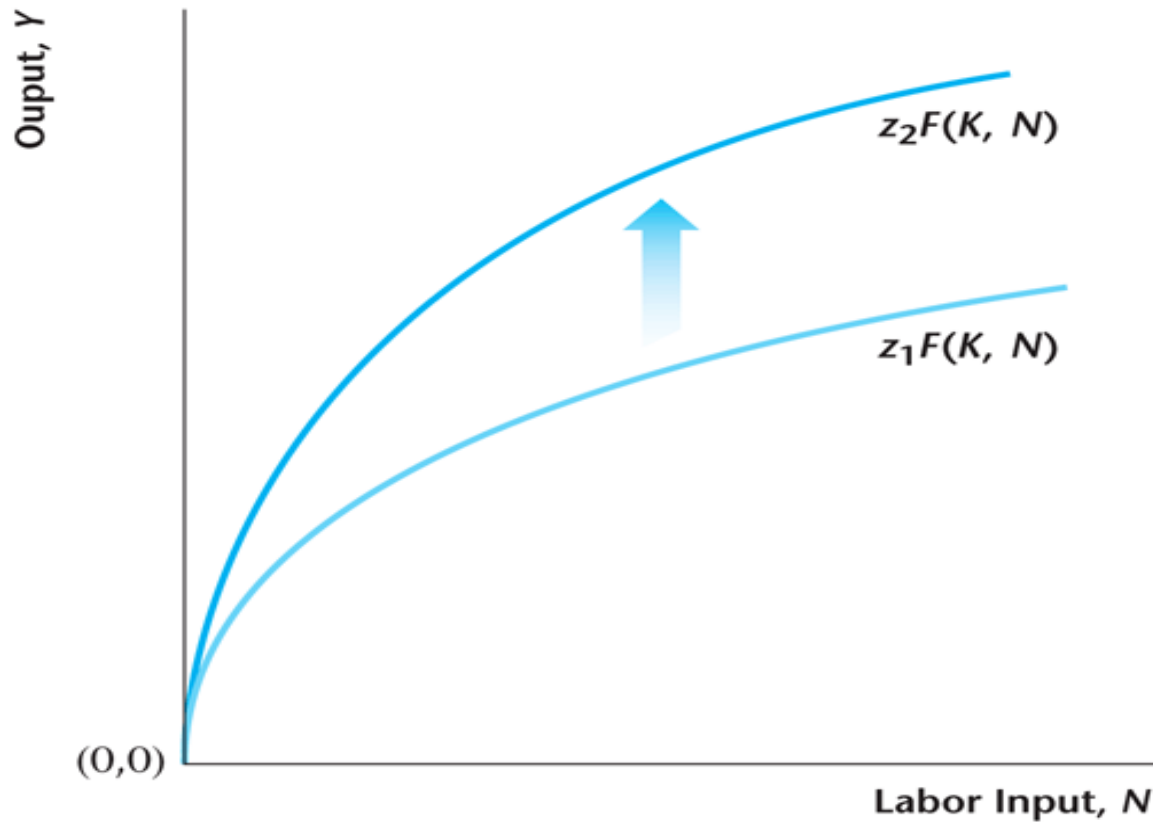


Figure 5.9

Competitive Equilibrium Effects of an Increase in Total Factor Productivity

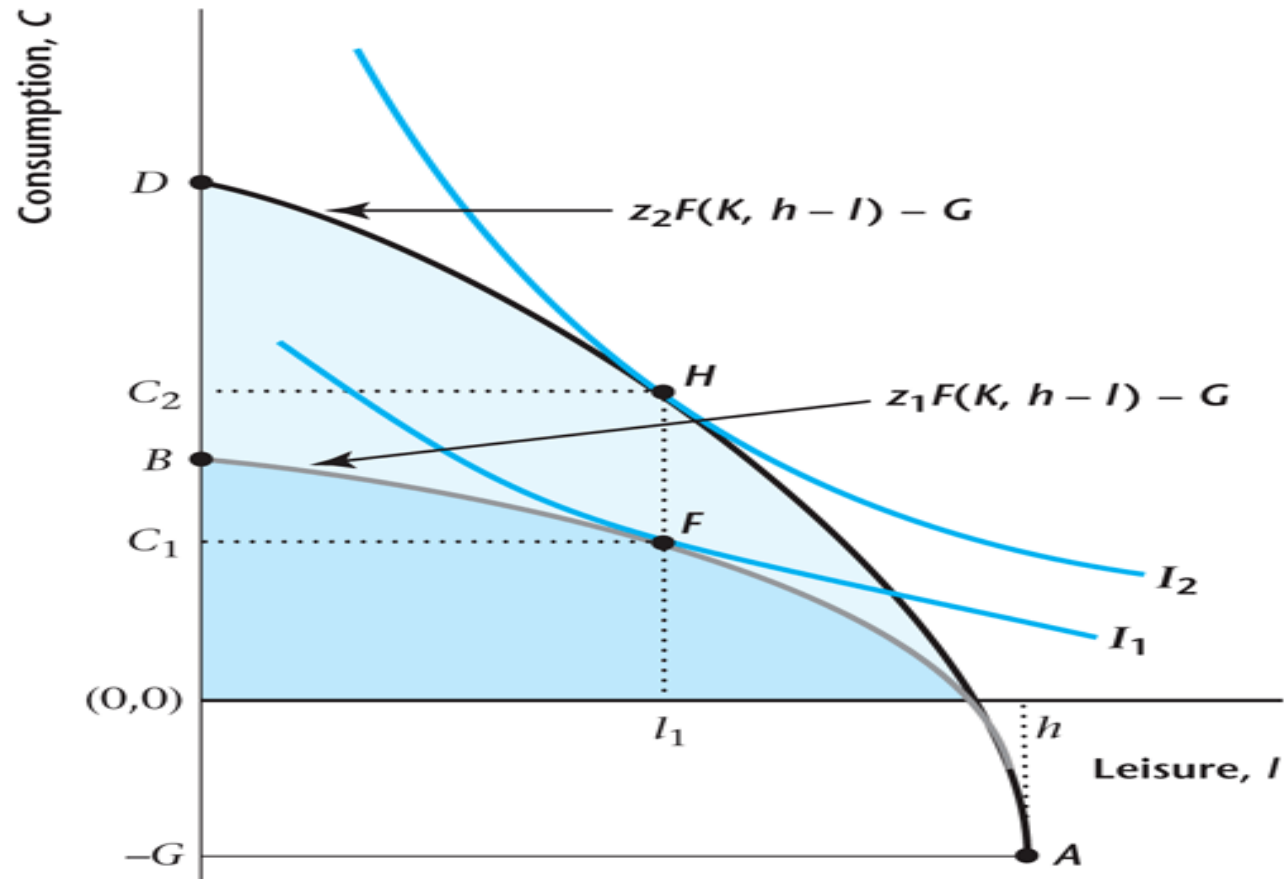
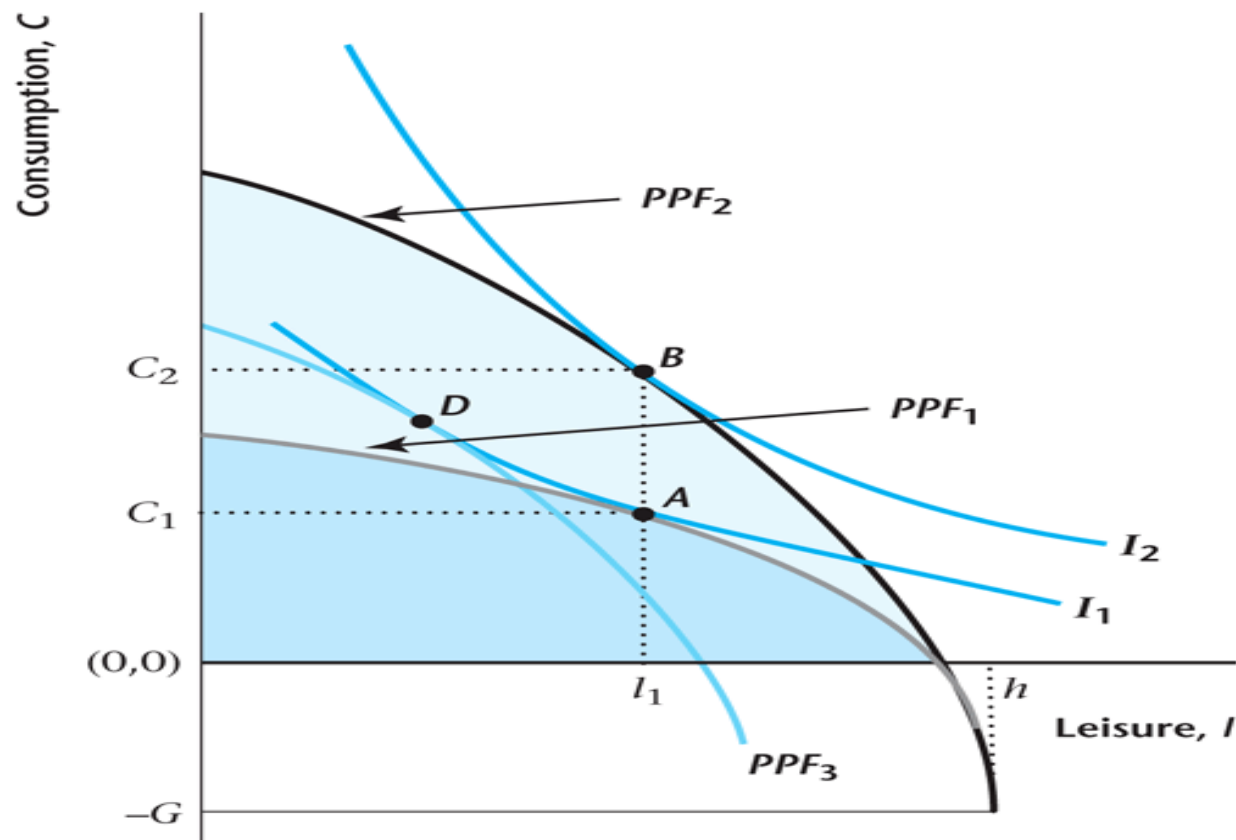


Figure 5.10

Income and Substitution Effects of an Increase in Total Factor Productivity



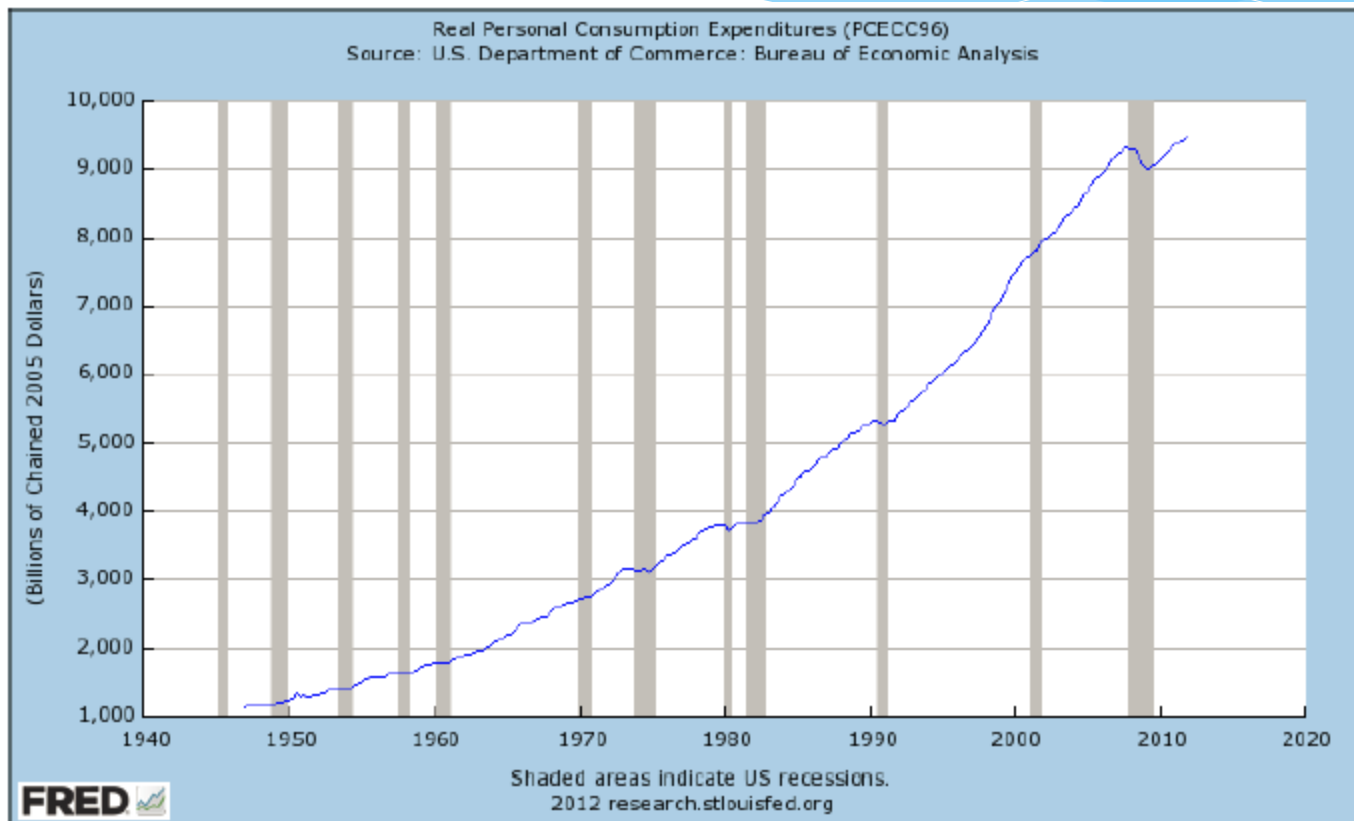
The effect of an increase in TFP

- C increase unambiguously
- Effect on N will depend on income and substitution effects
- w increases
- Y increases

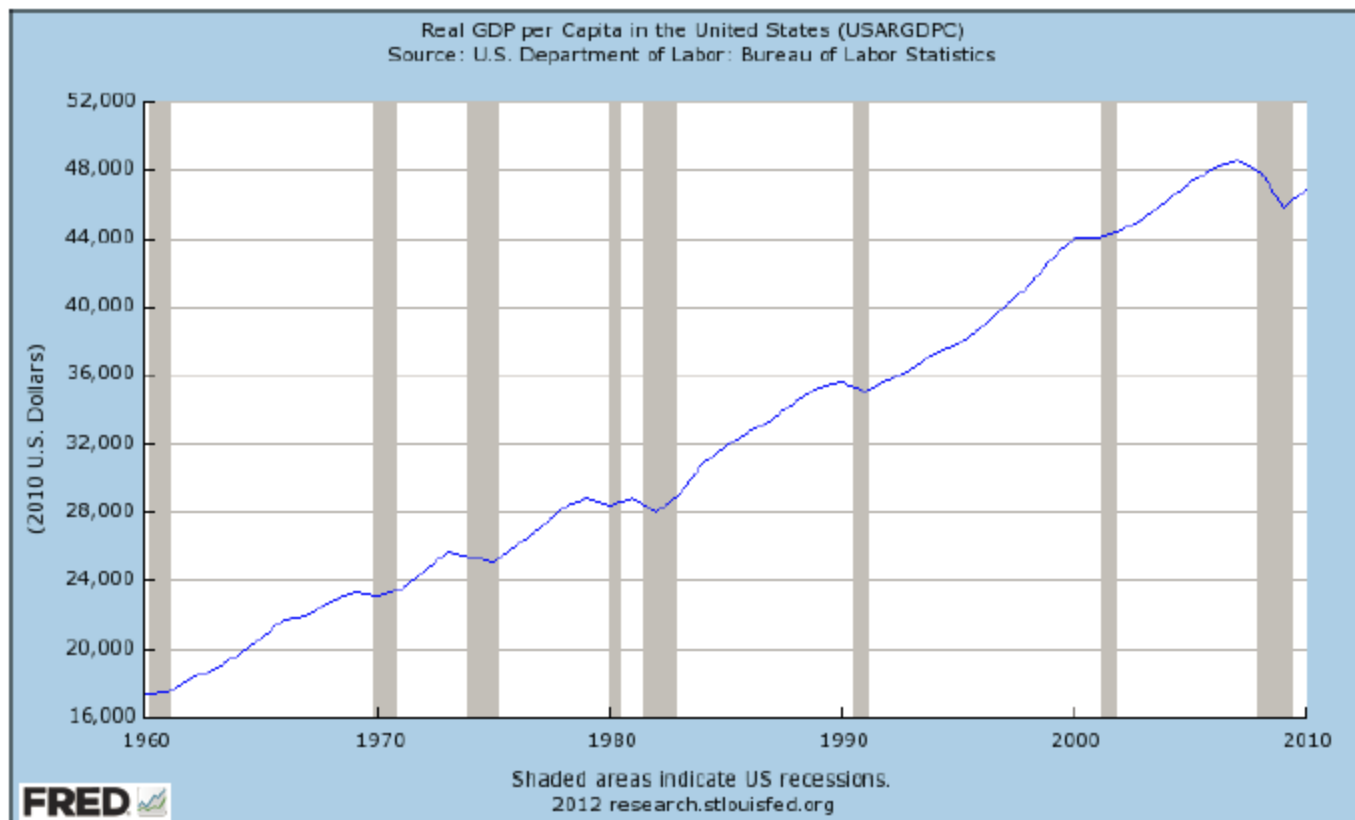
The effect of an increase in TFP

- Remember that z is *technological progress*.
- The model predicts that technological progress leads to more *consumption*, more *output*, higher *wages* and ambiguous effect on number of hours worked.
- We know that z has been increasing over time in the data. Let's see whether we observe the predicted trend in endogenous variables in the data.

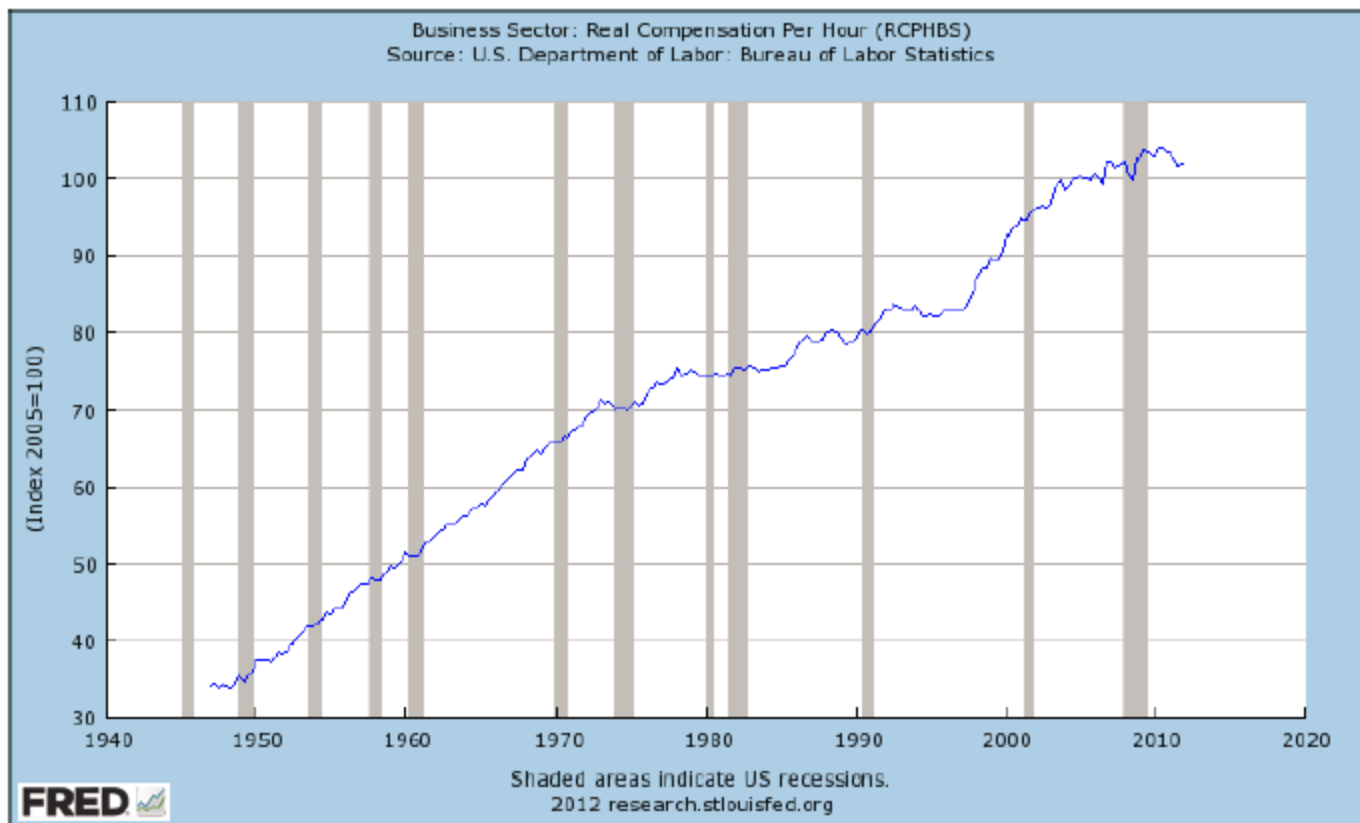
Consumption



Output



Compensation per hours



TFP

- So data seems consistent with this VERY simple model
- Really?

Let's Examine Changes in Hours Worked in the US since 1950.

- Hours worked per person have been roughly constant since WWII
 - overall: +7% from 1950 to 2000
- But: hours worked for men have fallen and market hours for women have increased

Hours worked per person

Year	Hours Worked per Person		
	Total Population	Sex	
		Males	Females
1950	22.34	34.18	10.87
1960	21.55	31.93	11.84
1970	21.15	29.75	13.38
1980	22.07	28.70	16.02
1990	23.86	29.11	19.03
2000	23.94	28.34	19.78
% Change 1950–2000	7.18	–17.09	81.98

TFP

$$\begin{aligned}\text{Hours per person} &= \text{Total Hours worked} / \text{Population} \\ &= \text{Total Hours worked} / \text{Labor Force} \times \\ &\quad \text{Labor Force} / \text{Population} \\ &= \text{Hours per worker} \times \text{Participation Rate}\end{aligned}$$

- Intensive Margin: Hours per worker
- Extensive Margin: Participation Rate

Hours worked per worker

Year	Hours Worked per Worker		
	Total Population	Sex	
		Males	Females
1950	42.40	44.00	38.19
1960	40.24	42.45	35.67
1970	38.83	41.33	34.70
1980	39.01	41.84	35.10
1990	39.74	42.59	36.30
2000	40.46	43.27	37.15
% Change 1950–2000	–4.56	–1.65	–2.73

Employment-to-population ratio

Year	Employment-to-Population Ratio, in %		
	Total Population	Sex	
		Males	Females
1950	52.69	77.69	28.46
1960	53.55	75.22	33.20
1970	54.47	71.98	38.55
1980	56.59	68.60	45.64
1990	60.04	68.35	52.42
2000	59.17	65.49	53.25
% Change 1950–2000	12.30	–15.70	87.10

Changes in Hours worked

- Hours worked per person have been roughly constant since WWII
 - overall: +7% from 1950 to 2000
- Hours per worker: -4.5% total population
- Large reallocation of working hours from males to females
 - Participation Rate of Men: -15%
 - Participation Rate of Women: +87%
- Improved technology in home sector (appliances, home delivery, the Internet, e-shopping).

Distortionary Tax

- Economist Arthur Laffer.
- Ronald Reagan's Economic advisor during 1980 U.S. election.
- Advice:
 - decreasing income tax rates,
 - will increase the quantity of labor supplied,
 - and increase tax revenues.

Distortionary Tax

- George H.W. Bush (Reagan's opponent in the Republican primaries):
 - This is “voodoo economics”
- 2001: tax cut proposals during George W. Bush's campaign for President
 - using more or less the same argument,
 - supposed to expire at the end of 2010,
 - two-year extension by Obama.

Production Function Without Capital

- Labor is the only input, but there is still constant returns to scale (linear production function).

$$Y = zN$$

Production Possibilities Frontier

$$C = z(h - l) - G$$

Consumer's Budget Constraint

$$C = w(1 - t)(h - l) + \pi,$$

Profits for the Firm

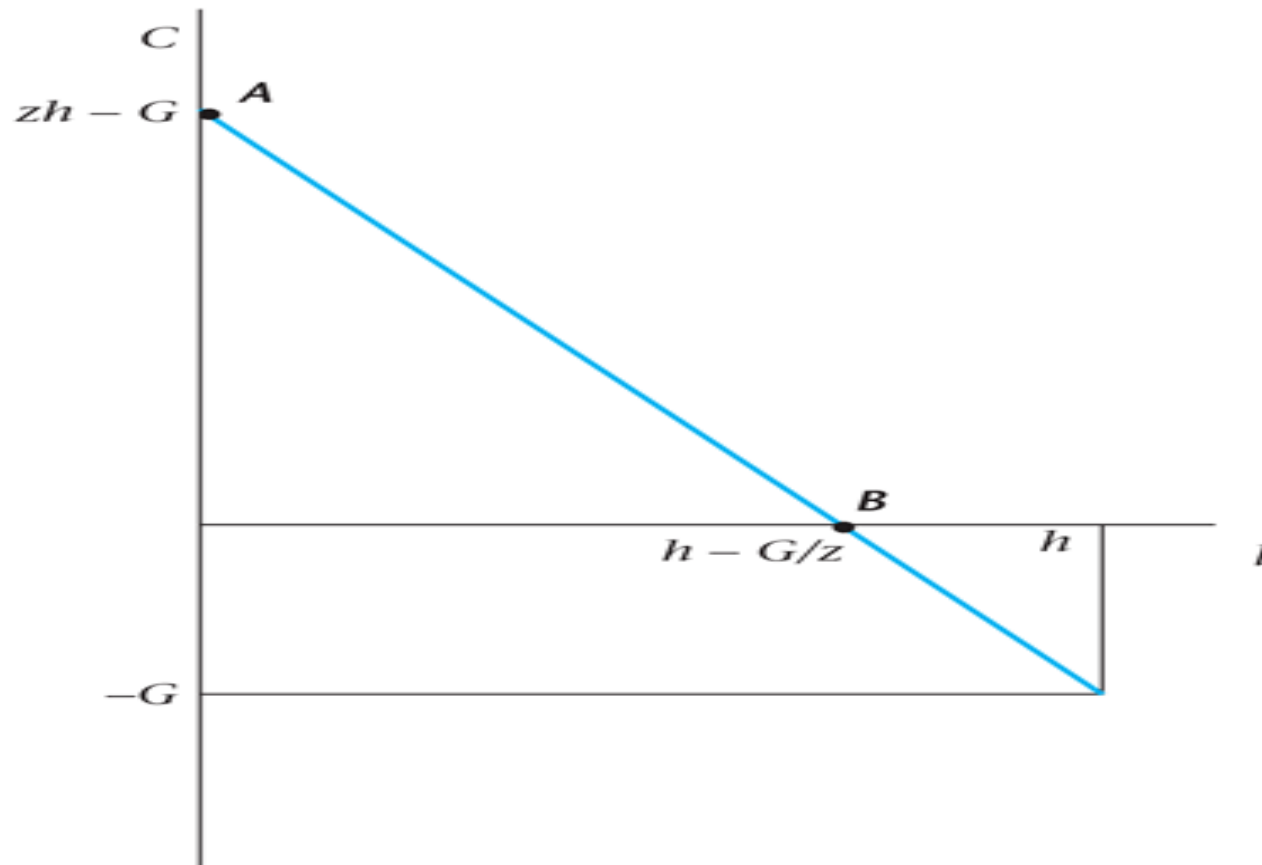
$$\pi = Y - wN^d = (z - w)N^d,$$

The Consumer's Budget Constraint in Equilibrium

$$C = z(1 - t)(h - l).$$

Figure 5.15

The Production Possibilities Frontier in the Simplified Model



Revenue for the Government Given the Tax Rate t

$$REV = tz[h - l(t)]$$

Figure 5.16

The Labor Demand Curve in the Simplified Model

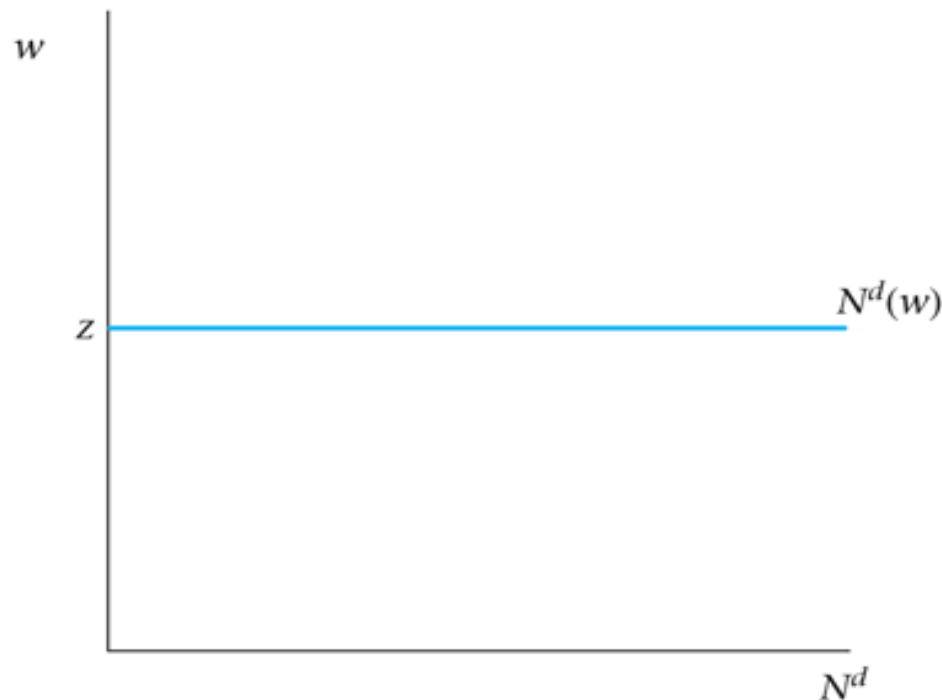


Figure 5.17

Competitive Equilibrium in the Simplified Model with a Proportional Tax on Labor Income

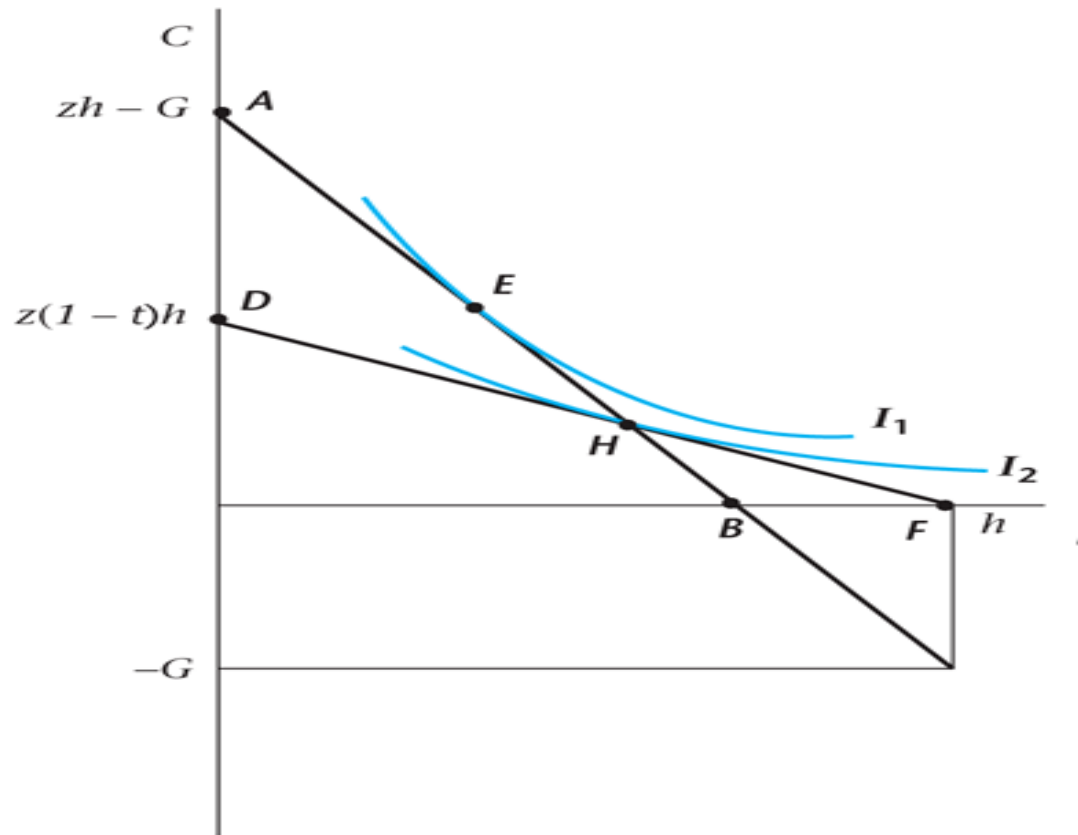


Figure 5.18

A Laffer Curve

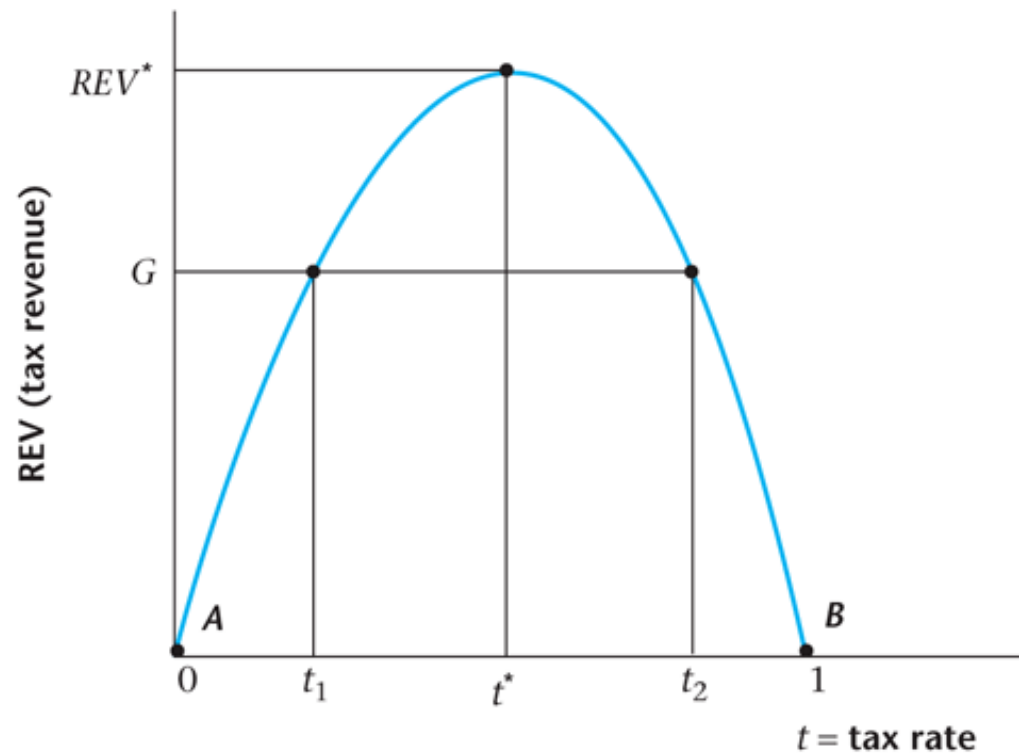
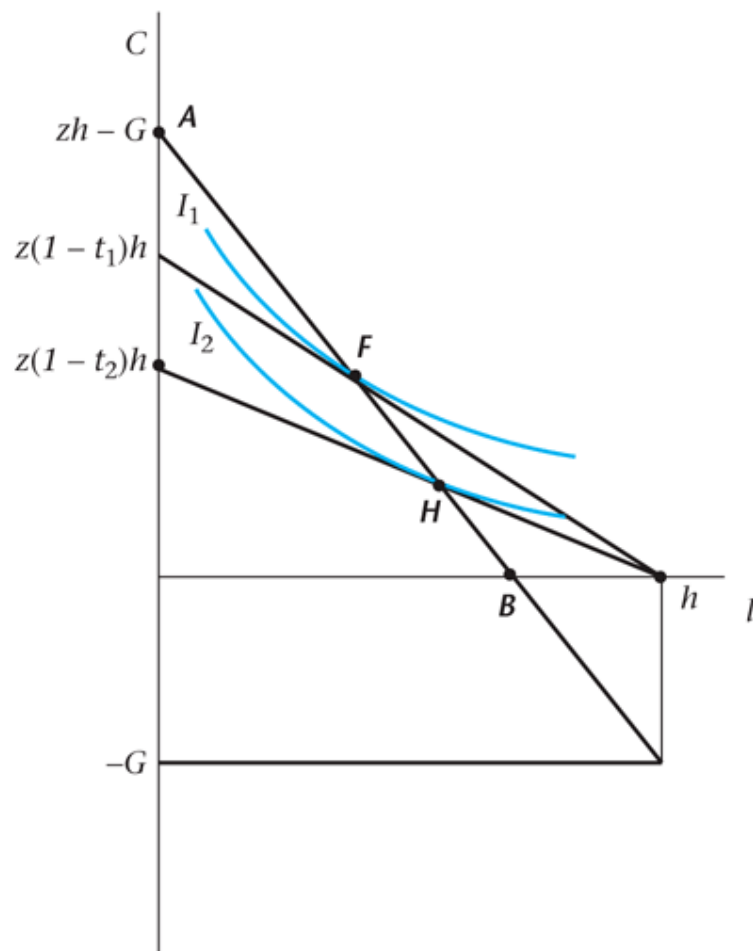


Figure 5.19

There Can Be Two Competitive Equilibria



Is the U.S. Economy on the Bad Side of the Laffer Curve?

- Reagan administration: cuts in marginal tax rates in 1981.
- Bush administration: cuts in marginal tax rates in 2001.
- Results: decrease in tax revenue in both cases.
- Implication: We are on the good side of the Laffer curve.

Figure 5.20

Federal Personal Taxes as a Percentage of GDP

